Predicting Income from Census Data: A Supervised Learning Proof-of-Concept

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Dataset used: Adult / Census Income (OpenML, `name="adult"`, version 2).

1 Problem Framing and Exploratory Data Analysis

1.1 Task and train/test split

The task is to predict whether an individual earns more than $50,000 per year from 14 demographic and employment attributes of the 1994 US Census [1]. The positive class (>50K) represents 23.9% of the observations, so the problem is imbalanced (Section 3).

The data are split once into a training set (80%, 39,032 rows) and a test set (20%, 9,758 rows), with stratification on the target and a fixed seed; both sets contain 23.9% of positives. The test set is not used before the final evaluation: model comparisons rely on 5-fold stratified cross-validation within the training set, and every relationship between a feature and the target reported below is computed on the training set only.

1.2 Data quality

Duplicates and removed columns. The raw data contain 48,842 rows, of which 52 are exact duplicates; these are removed, leaving 48,790 observations. This is done before dropping any column: if `fnlwgt` were removed first, 6,374 rows would appear duplicated, although they are distinct individuals who share the same profile, not recording errors. Two columns are then removed. `fnlwgt` is a sampling weight indicating how many people in the US population a respondent represents; it is not linked to the individual's characteristics. `education_num` maps one-to-one onto `education` and is therefore redundant.

Missing values. Three columns contain missing values: `workclass` (5.7%), `occupation` (5.7%) and `native_country` (1.8%). 10 rows with no occupation belong to the `Never-worked` class, so they have no occupation by definition. Then we have 2,795 rows where both `workclass` and `occupation` are missing together.

To choose how to handle them, we need to know whether these values are missing at random [2]. We cannot know why a value is missing, but we can compare people with and without a reported work class (Table 1).

Table 1: Profile of people with a missing or reported work class (training set).

	Work class missing	Work class reported
Aged 20 or younger, or 65 or older	39.2%	9.9%
Works less than 35 hours per week	44.7%	15.6%
Lives with parents (`Own-child`)	30.4%	14.7%
Earns >50K	9.4%	24.8%
Earns >50K, aged 21 to 64 only	10.9%	26.7%

People with a missing work class are much more often very young or old, work part time more often, and more often live with their parents. This profile suggests people outside regular employment, such as students and retirees. As a result, they earn >50K much less often (9.4% against 24.8%). Even among people aged 21 to 64, the gap remains (10.9% against 26.7%), so age alone does not explain it. The missing values of `workclass` and `occupation` are therefore not random: their absence is probably linked to the person's situation and to income.

The case of `native_country` is different. People with a missing country earn >50K in 25.1% of cases, close to the overall rate of 23.9%. Their absence does not seem linked to the person's situation.

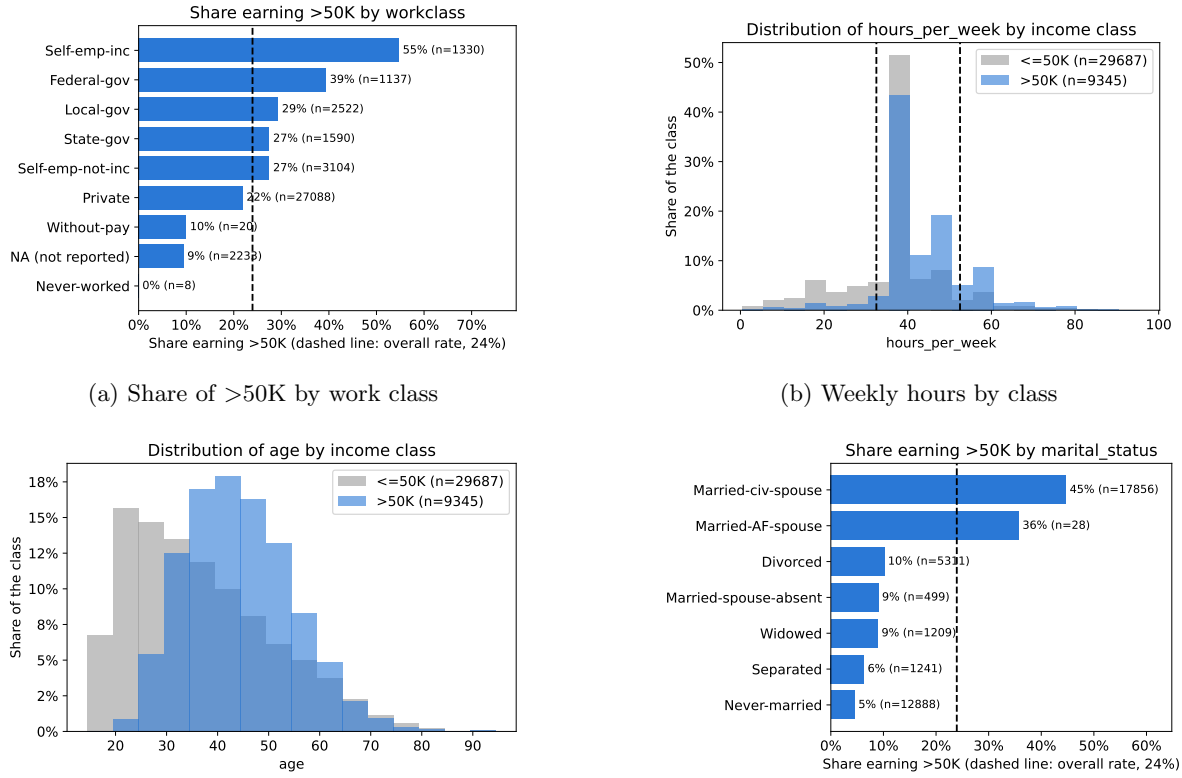
Extreme values. We examined the four numerical variables in two ways: a standard outlier rule and the maximum values.

First, we applied Tukey's rule that flags values below $Q_1 - 1.5\,\text{IQR}$ or above $Q_3 + 1.5\,\text{IQR}$. For age, the rule flags 0.4% of people, all older than 78. These are real elderly people, not errors. For weekly hours, it flags 27.6% of people. This happens because 46.7% of people work exactly 40 hours, so $Q_1 = 40$, $Q_3 = 45$, and the bounds are 32.5 and 52.5 hours. As Figure 1b shows, the rule then flags wrongly normal part-time jobs and long working weeks. Regarding capital income, most values are zero (91.7% for gains and 95.3% for losses). Therefore, $Q_1 = Q_3 = 0$ and every non-zero value would be flagged. The non-zero values are also very skewed (handled in Section 2.2): the median gain is 7,298, but the maximum is 99,999 (Appendix A).

Second, the maximum values show that some variables are capped. 54 people are exactly 90 years old, against 6 and 2 for ages 88 and 89. 137 people work exactly 99 hours per week. 244 people have a capital gain of exactly 99,999, while the next highest gain is 41,310. These values probably mean "this amount or more" and are not errors. For instance, the 99,999 value is very informative: 100% of these people earn >50K.

Rare and inconsistent categories. Some categories contain very few people. In the training set, `Never-worked` has 8 people, `Without-pay` 20 and `Armed-Forces` 12. In the full dataset, 17 countries have fewer than 50 people. Rates computed on such small groups are unreliable. We also found four inconsistent rows: one woman is recorded as a `Husband` and three men as a `Wife`.

1.3 Relationships with the target



(a) Share of >50K by work class

(b) Weekly hours by class

(c) Age by class

(d) Share of >50K by marital status

Figure 1: Exploratory analysis on the training set. (a) and (d): share of people earning >50K in each category; the dashed line is the overall rate. In (a), missing work classes are shown as a separate bar. (b) and (c): distribution within each class (each class sums to 100%). In (b), the dashed lines are Tukey's bounds: $40 - 1.5 \times 5 = 32.5$ and $45 + 1.5 \times 5 = 52.5$ hours.

Employment. Income varies strongly with the type of employment (Figure 1a). The private sector covers 69% of people, and 22% of them earn >50K. The rate is higher for federal employees (39%) and highest for self-employed people with their own company (55%). People with a missing work class have a low rate (9%), in line with the profile of Table 1. Working time also matters (Figure 1b): 40.2% of people who work more than 40 hours earn >50K, against 9.4% of people who work less than 40 hours.

Demographics. High earners are older: their median age is 43, against 34 for the others (Figure 1c), and almost no one under 25 earns >50K (1.2%). Marital status also matters. Married people living with their spouse earn >50K in 44.7% of cases, while the other frequent statuses stay between 4.5% and 10.1% (Figure 1d). Education shows a similar gap, from 1.6% (`Preschool`) to 74.1% (`Doctorate`). Finally, 21.3% of high earners have a capital gain, against 4.1% of the others.

Sex. The share of high earners differs by sex: 30.3% for men against 11.1% for women. In addition, `relationship` is closely linked to sex: a `Husband` is a man in all cases but one, and a `Wife` is a woman in all cases but three.

2 Data Preparation and Feature Engineering

2.1 Missing values

Section 1.2 showed that the missing values of `workclass` and `occupation` are not random. We considered three options. Deleting these rows would remove 5.7% of the data and, above all, a specific group of low earners, which would bias the model. Replacing them with the most frequent value (`Private`) would mix two groups with different incomes (9.4% against 21.9%). We therefore keep them as a separate category, called `Missing`, in both columns. The model can then learn what the absence means.

For `native_country`, the share of high earners (25.1%) is similar to the overall share (23.9%), which suggests that this missingness is unrelated to the target, and it concerns only 1.8% of rows. We therefore replace these values with the most frequent country, the United States, which covers 91.4% of people whose country is known. It is the most likely value, so the risk of error is small.

Both treatments are applied inside the pipeline: the most frequent value is computed on the training folds only, so no information comes from the data used for evaluation.

2.2 Extreme values

We do not remove any extreme value, for two reasons: none of them is an error and removing them would lose information, especially the capital gain of 99,999, which always means >50K.

The remaining question is whether these values can harm the models. Tree-based models are not affected, because they split the data with thresholds. The logistic regression is more sensitive so for this model, numerical features are standardised and capital income is log-transformed (Section 2.5), which reduces the weight of very large values.

2.3 Rare and inconsistent categories

Regarding the rare categories, we keep them. Most of them disappear in the engineered features of Section 2.5: rare countries are grouped in `native_us`, and `Married-AF-spouse` is merged into `is_married`. The others, such as `Never-worked`, are too small to influence the models. Regarding the four inconsistent rows, we also keep them. They represent less than 0.01% of the data and cannot affect the results, while correcting them would require guessing which value is wrong.

2.4 Encoding and scaling

Models need numerical inputs. The six categorical variables have no natural order, so we use one-hot encoding: each category becomes a 0/1 column. This concerns `workclass`, `education`, `occupation`, `relationship`, `race` and `sex`. Very rare categories not seen during training are ignored in the encoder. We keep one column per category. Although this creates one redundant column per variable, it is not a problem: the logistic regression uses L2 regularisation by default, and trees are not affected. Numerical features are standardised. All models use the same pipeline, so that Section 4 compares the models only.

2.5 Engineered features

We create three features, each replacing raw columns (Table 2).

Reducing `native_country` to `native_us` may seem to discard information: for instance, 5.1% of people born in Mexico earn >50K, against 42.9% of people born in India. However, 17 countries have fewer than 50 people, so their rates are unreliable. Moreover, across the 24 countries with at least 50 people, the share of high earners and the share of university graduates have a correlation of 0.87 (an association between country averages, not a causal link). Since education is already in the model, a detailed country variable would add little. The indicator keeps the main difference: 24.4% of people born in the United States earn >50K, against 18.4% of the others.

Table 2: Engineered features and the columns they replace.

Feature	Replaces	Definition	Justification
`capital_net`	`capital_gain`, `capital_loss`	$\text{sign}(g-\ell)\log(1+\lvert g-\ell\rvert)$, with g the gain and ℓ the loss	No one has both a gain and a loss, so one column keeps all the information. The log reduces the skew (Appendix A).
`is_married`	`marital_status`	1 if married and living with spouse	The income gap is concentrated in this group: 44.7% against at most 10.1% for the other frequent statuses.
`native_us`	`native_country`	1 if born in the United States	Most countries are too small for reliable rates, and the gaps between countries mostly follow education (see below).

3 Class Imbalance

3.1 Metrics

The positive class represents only 23.9% of the data. With such an imbalance, accuracy is misleading: a model that always predicts ≤50K reaches 76.1% accuracy without finding a single high earner. We therefore do not use accuracy, and choose metrics that focus on the positive class.

Our main metric is the area under the precision–recall curve (PR-AUC). It measures how well the model ranks high earners above the others, over all possible thresholds. It focuses on the positive class, and a random model scores 0.239, the share of positives [3]. We also report ROC-AUC, the standard ranking metric, but only as a secondary metric. ROC-AUC relies on the false positive rate, FP/(FP + TN). Since true negatives are numerous, this rate stays low even when the model makes many false alarms, so ROC-AUC can look better than the model really is. Our imbalance is moderate, so this effect is smaller than in extreme cases, but it remains visible: in Section 4, ROC-AUC is always much higher than PR-AUC.

Ranking metrics do not show the decisions the model actually makes. We therefore also report three metrics at the default threshold of 0.5: recall, the share of high earners that the model finds; precision, the share of predicted high earners that are correct; and F1, which balances the two.

3.2 Chosen remedy

We handle the imbalance with class weighting. This method changes the training loss: errors on high earners count more, so the model cannot ignore them. It does not change the data and works inside the pipeline without any risk of leakage.

In the code, the weight is passed to each model as `class_weight={0: 1, 1: w}`: an error on a high earner counts w times more than an error on a low earner. This is equivalent to the `scale_pos_weight` parameter seen in class, but `class_weight` is available in all three of our models.

The best value of w is not known in advance. We therefore treat it as a hyperparameter and tune it with Optuna (Section 6), between 1 and $n_0/n_1 \approx 3.18$, where n_0 and n_1 are the numbers of negatives and positives in the training set. The lower bound, $w = 1$, means no weighting: it leaves Optuna free to conclude that weighting is not useful.

4 Modelling Progression

4.1 Evaluation protocol

The goal of this section is to choose the model family, not its final settings. All models therefore keep their default hyperparameters. Each model is placed in the same pipeline (Section 5) and evaluated with 5-fold stratified cross-validation on the training set. The training set is split into five parts that keep the class ratio. Each part is used once for evaluation, while the model is trained on the other four. We report the mean score over the five folds. To detect overfitting, we also compute the PR-AUC on the training folds.

4.2 Models

We compare three models of increasing flexibility.

Baseline: logistic regression. It is a simple linear model, fast and easy to interpret. It sets the minimum level that more complex models must beat.

Random forest. It averages many decision trees, each trained on a random sample of the data. Trees can capture non-linear effects and interactions between features, and averaging them reduces the instability of a single tree.

Gradient boosting with LightGBM. It builds trees one after another, each new tree correcting the errors of the previous ones. Among boosting libraries, we chose LightGBM over XGBoost and CatBoost. Our dataset is of moderate size and we have no strong resource constraint, so raw performance does not clearly separate these libraries. Essentially, we chose LightGBM because it was covered in class and because it accepts the same `class_weight` parameter as the scikit-learn models, so that the imbalance is handled identically for all models.

4.3 Results

Table 3: Mean scores over 5-fold cross-validation on the training set (default hyperparameters, threshold 0.5). Train PR-AUC is computed on the training folds.

	Log. reg.	Rand. forest	LightGBM
Precision	0.700	0.695	**0.781**
Recall	0.566	0.618	**0.658**
F1	0.626	0.654	**0.714**
ROC-AUC	0.890	0.889	**0.929**
PR-AUC	0.709	0.731	**0.830**
Train PR-AUC	0.711	0.991	0.860

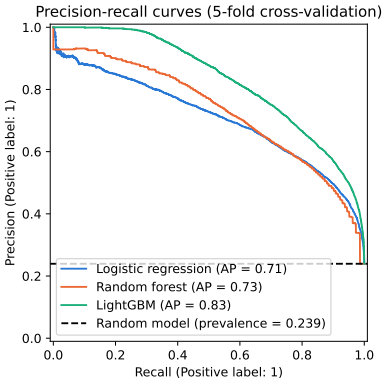


Figure 2: Precision–recall curves from out-of-fold predictions (5-fold cross-validation on the training set). AP is the PR-AUC; the dashed line is a random model.

Ranking of the models. LightGBM is clearly the best model (Table 3). Its PR-AUC (0.830) is 12.1 points higher than the baseline, and it also has the best ROC-AUC, precision, recall and F1. Figure 2 confirms this: its curve lies above the two others at almost every level of recall. The random forest improves on the baseline only slightly (+2.2 points of PR-AUC), and the two models have the same ROC-AUC (0.889 and 0.890). Their curves are close, and they meet at high recall. All models are far above the random model (0.239), so they all learn a real signal.

Overfitting. The gap between training and cross-validated PR-AUC differs strongly between models. The logistic regression shows almost no gap (0.711 against 0.709). It does not overfit, but its simple linear form limits its score. The random forest shows a very large gap (0.991 against 0.731). With default settings, its trees grow until each leaf is almost pure, so the model learns the training data almost by heart. This overfitting probably explains why it barely beats the baseline despite its flexibility. LightGBM shows a moderate gap (0.860 against 0.830). By default, its trees are limited to 31 leaves and each tree only corrects part of the error, which keeps its complexity under control. LightGBM is also the most stable model across folds: the standard deviation of its PR-AUC is 0.002, against 0.006 for the random forest and 0.009 for the logistic regression.

Conclusion. LightGBM is our main candidate. All three models are nevertheless tuned in Section 6, so that each model family is compared at its best. The random forest in particular should benefit from tuning: its gap between training and cross-validation suggests that limiting the growth of its trees could improve it.

5 Pipeline

All preparation steps and the model are wrapped in a single scikit-learn `Pipeline` (function `build_pipeline` in `src/pipeline.py`). The pipeline takes the raw data as input and returns a prediction. It chains the three steps of Table 4.

Using a single pipeline has three benefits: it prevents data leakage, the three models receive exactly the same input, and it is a single object that can be saved and used to serve predictions on raw data.

Table 4: Steps of the pipeline.

Step	What it does	Learned from the data
1. Feature engineering	Creates `capital_net`, `is_married` and `native_us`, and removes the columns they replace (Section 2.5)	Nothing
2. Preprocessing	Numerical features: imputation with the most frequent value, then standardisation. Categorical features: `Missing` category, then one-hot encoding	Most frequent values, means, standard deviations, list of categories
3. Model	Logistic regression, random forest or LightGBM	The model parameters

6 Hyperparameter Tuning

6.1 Method

We tune the three models with Optuna (`src/tune.py`). For each model, Optuna runs a study of 30 trials. Each trial proposes a set of hyperparameters, and the corresponding pipeline is scored with the same 5-fold stratified cross-validation as in Section 4, on the training set only. The objective is the mean cross-validated PR-AUC, our main metric (Section 3.1). Optuna uses its default sampler (TPE), which proposes new values close to the best trials found so far. We fix its random seed, so the study gives the same results at every run.

Table 5 gives the search space. Each range starts from the default value of the hyperparameter and extends in the direction suggested by Section 4. We also carefully selected the hyperparameters to avoid redundancy: for instance, `min_samples_leaf` and `min_samples_split` both limit the growth of the trees, so we tune only the first one.

Table 5: Search space of each model. The default values are those of Section 4.

Model	Hyperparameter	Default	Range	Justification of the range
All	w	1	1 to 3.18	From no weighting to full rebalancing (n_0/n_1)
Log. reg.	`C` (log scale)	1	0.001 to 10	From 1000 times smaller to 10 times larger than the default.
Random forest	`n_estimators`	100	100 to 500	From the default to 5 times the default (more trees do not cause overfitting)
	`max_depth`	None (no limit)	5 to 30	Limits the depth of the trees, which is unlimited by default and leads to overfitting (Section 4)
	`min_samples_leaf`	1	1 to 20	Only larger leaves than the default, for the same reason
	`max_features`	$\sqrt{58} \approx 7$	4 to 28	From about half to 4 times the default (58 features after encoding)
LightGBM	`n_estimators`	100	50 to 1000	From half to 10 times the default: a lower learning rate needs more trees
	`num_leaves`	31	7 to 63	From a quarter to twice the default
	`learning_rate` (log scale)	0.1	0.01 to 0.2	From 10 times smaller to twice the default

6.2 Results

Table 6 compares the default and tuned versions of each model (see also Figure 8 in Appendix A). For every model, the best values lie strictly inside their ranges, so the ranges did not limit the search. Figure 9 (Appendix A) shows the score of every trial.

Logistic regression. Tuning brings no gain: the PR-AUC stays at 0.709. Optuna also selects almost no weighting ($w = 1.02$) and a weak regularisation. This confirms the diagnosis of Section 4: the model does not overfit, and its score is limited by its linear form.

Table 6: Default and tuned models (5-fold cross-validation on the training set, threshold 0.5). Default scores come from Table 3.

Model	Version	Prec.	Recall	F1	PR-AUC	Train PR-AUC	Best hyperparameters
Log. regression	Default	0.700	0.566	0.626	0.709	0.711	
	Tuned	0.699	0.571	0.628	0.709	0.711	$w = 1.02$, `C` $= 4.05$
Random forest	Default	0.695	0.618	0.654	0.731	0.991	
	Tuned	0.666	0.761	0.710	0.809	0.886	$w = 2.05$, 400 trees, depth 18, 3 per leaf, 25 features
LightGBM	Default	0.781	0.658	0.714	0.830	0.860	
	Tuned	0.750	0.702	**0.725**	**0.832**	0.861	$w = 1.31$, 550 trees, 9 leaves, rate 0.083

Random forest. The random forest benefits most from tuning: its PR-AUC rises from 0.731 to 0.809 (Table 6). The gap between training and cross-validation falls from 26 points (0.991 against 0.731) to 8 points (0.886 against 0.809). The tuned trees are shallower (depth 18) and keep at least 3 observations per leaf, so they no longer learn individual training rows, as expected in Section 4. Optuna also selects 25 features per split, far above the default of 7.

LightGBM. The PR-AUC barely changes (0.830 to 0.832): the default settings were already close to the best ones. The tuned model uses very small trees (9 leaves) with many boosting steps, and its gap between training and cross-validation stays at about 3 points (0.861 against 0.832). The main change is at the default threshold: with $w = 1.31$, recall rises from 0.658 to 0.702, while precision falls from 0.781 to 0.750, and F1 improves slightly.

Role of the class weight. Optuna selects weights between 1.02 and 2.05, below full rebalancing (3.18). Table 6 and Figure 8 (Appendix A) show the effect of w: for the random forest and LightGBM, recall rises and precision falls after tuning.

Number of trials. Figure 9 (Appendix A) shows that the scores level off quickly. For the logistic regression and LightGBM, the trials after the first 15 improve the best PR-AUC by less than 0.001. The random forest still improves slightly after 15 trials (+0.002), and its best trial is one of the last ones. More trials could therefore add a little to its score, but this gain is small compared with its gap to LightGBM (2.3 points). 30 trials are therefore sufficient to compare the three models.

Choice of the model. LightGBM remains the best model after tuning (PR-AUC of 0.832, against 0.809 for the random forest and 0.709 for the logistic regression), so it is our final model. Note that the best cross-validated score is slightly optimistic: the best of 30 trials is chosen and scored on the same folds. The unbiased estimate of the final model will come from the untouched test set (Section 9).

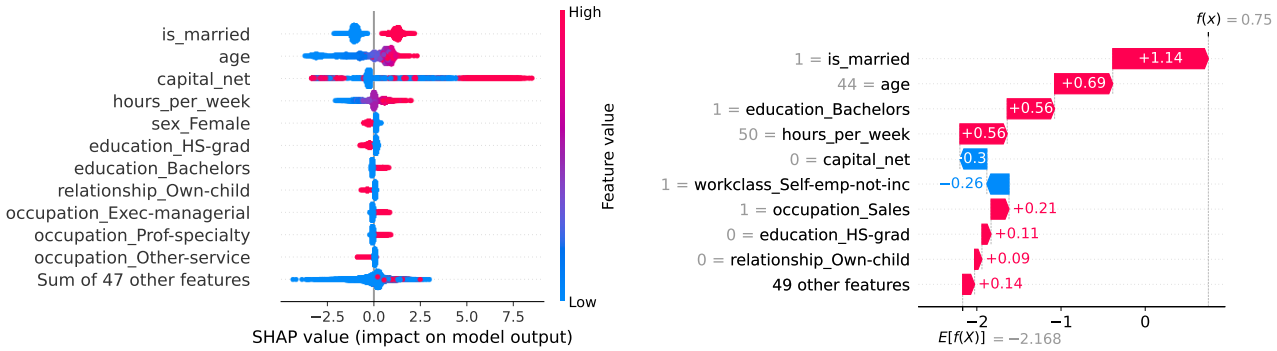
7 Experiment Tracking

All tuning runs are logged with MLflow, in a local database (`mlflow.db`). The experiment contains 94 runs, organised as follows:

- **One run per model** (3 runs). It stores the best hyperparameters and the cross-validated metrics of the best trial, and the file `best_params.json` as an artifact.
- **One run per trial** (90 runs), nested in the run of its model. Each one stores the hyperparameters of the trial, including w, and all the metrics of Section 3.1: precision, recall, F1, ROC-AUC and PR-AUC in cross-validation, the standard deviation of PR-AUC across folds, and the training PR-AUC.
- **One summary run**. It stores, as artifacts, the comparison table (`tuning_results.csv`), the best hyperparameters of the three models, and Figures 8 and 9.

8 Explainability with SHAP

We explain the final model with SHAP (`src/explain.py`). The tuned LightGBM is fitted on the whole training set, and SHAP values are computed for all 39,032 people of the training set with `TreeExplainer`. For each person, SHAP splits the model output into one contribution per feature, added to a baseline (the average output). For LightGBM, these contributions are on the log-odds scale, not on the probability scale. Since SHAP describes the model and does not evaluate it, the test set is not needed.



(a) Global view: one point per person.

(b) Local view: one person.

Figure 3: SHAP values of the final model on the training set (log-odds scale). In (a), the colour gives the value of the feature, and the horizontal position its effect on the prediction.

Global explanation. Figure 3a ranks the features by their mean absolute SHAP value. The four numerical features dominate: `is_married` (1.12), `age` (0.88), `capital_net` (0.73) and `hours_per_week` (0.37). The directions are consistent with Section 1. Being married pushes the prediction up, and not being married pushes it down. Older people and people who work more hours receive higher predictions. Large capital gains have the strongest individual effects, up to about +8 in log-odds, but they concern few people. Among the one-hot columns, being a woman (`sex_Female`, 0.18) and having only a high-school diploma (`education_HS-grad`, 0.17) push the prediction down. The model therefore uses sex directly, which matters for fairness. Note that each categorical feature is split into several one-hot columns, so the total effect of `education` or `occupation` is spread over several rows of the figure.

Local explanation. Figure 3b explains one prediction: the first high earner of the training set that the model predicts correctly. This person is a 44-year-old married man with a bachelor's degree, self-employed in sales, who works 50 hours per week and has no capital income. The baseline, i.e. the average output of the model over the training set, is −2.17 in log-odds. Being married (+1.14), his age (+0.69), his degree (+0.56) and his working hours (+0.56) raise the prediction. The absence of capital income (−0.32) and his type of employment (−0.26) lower it. The final output is 0.75 in log-odds, which corresponds to a predicted probability of 0.68.

Limits. SHAP explains the fitted model, not the real world: a positive contribution of `is_married` does not mean that getting married causes a higher income.

9 Calibration and Final Evaluation

9.1 Method

A good PR-AUC means that the model ranks people well. It does not mean that a predicted probability of 0.3 corresponds to a 30% chance of earning >50K. We check this with three tools: the reliability diagram, the Brier score and the expected calibration error. For the Brier score and the ECE, lower is better.

We compare three versions of the final model (`src/calibration.py`): without calibration, with Platt scaling and with isotonic regression. In both cases, the calibrator is learned on out-of-fold predictions, i.e. on predictions for people the model did not train on. The three versions are compared with the same 5-fold cross-validation as before, on the training set only.

9.2 Results

Table 7: Calibration of the final model (5-fold cross-validation on the training set). The observed share of >50K is 0.239.

Version	Brier	ECE	Mean predicted prob.	PR-AUC	Precision	Recall
Uncalibrated	0.0883	0.0229	0.262	0.832	0.750	0.702
Platt (sigmoid)	**0.0873**	0.0033	0.240	0.832	0.787	0.652
Isotonic	0.0874	**0.0029**	0.240	0.830	0.782	0.660

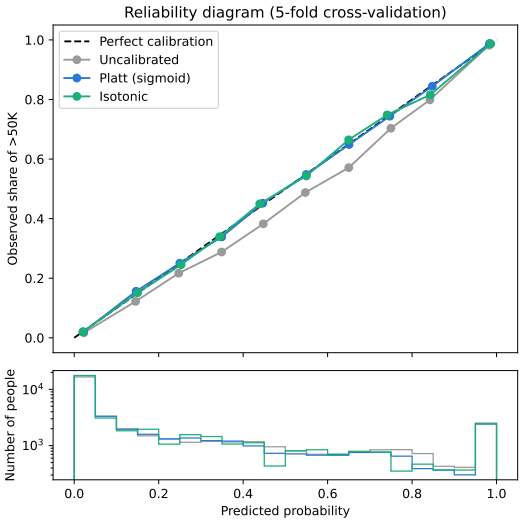


Figure 4: Reliability diagram (top) and distribution of the predicted probabilities (bottom, log scale), 5-fold cross-validation.

Diagnosis. Without calibration, the model overestimates the probability of earning >50K. Its mean predicted probability is 0.262, while the observed share is 0.239, and its curve lies below the diagonal in Figure 4. This is the expected effect of the class weight: with $w = 1.31$, errors on high earners cost more during training, which pushes the predictions upwards.

Choice of the method. Both methods bring the curve onto the diagonal and the mean prediction to 0.240. The ECE falls from 0.023 to about 0.003. Platt scaling has the lowest Brier score (0.0873). For simplicity and because results are very close, we keep Platt scaling.

Effect on decisions. Calibration lowers the probabilities but keeps the order of people, so the PR-AUC does not change (0.832). At the threshold of 0.5, however, some people now fall just below the threshold: recall falls from 0.702 to 0.652, and precision rises from 0.750 to 0.787. This is not a loss of quality: because of w, the uncalibrated model predicted >50K too often. After calibration, the threshold of 0.5 simply means that earning >50K is more likely than not.

9.3 Final evaluation on the test set

The test set is used here for the first time (`src/evaluate.py`), and from now on only to measure: nothing is chosen from it. The final LightGBM is fitted on the whole training set, with and without Platt scaling, and both versions are scored once. Nothing is changed after this step.

Table 8: Final evaluation on the untouched test set (9,758 people, threshold 0.5).

	Uncalibrated	Platt (final)
PR-AUC	0.822	0.822
ROC-AUC	0.926	0.926
Precision	0.730	0.771
Recall	0.691	0.639
F1	0.710	0.699
Brier score	0.0913	0.0900
ECE	0.0261	0.0059
Mean predicted prob.	0.266	0.242

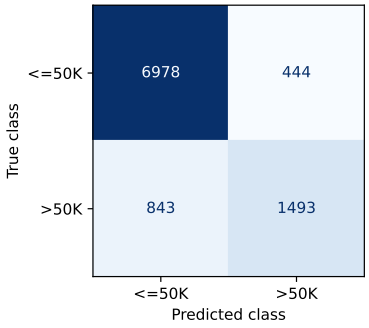


Figure 5: Confusion matrix of the final model on the test set (threshold 0.5).

The final model reaches a PR-AUC of 0.822 on the test set (Table 8), against 0.832 in cross-validation. This small drop of one point is expected: the cross-validated score was slightly optimistic, because it was the best of 30 trials (Section 6). The test confirms the results of the calibration step: Platt scaling reduces the ECE from 0.026 to 0.006 and brings the mean predicted probability (0.242) close to the observed share (0.239), without changing the PR-AUC. Figure 5 shows the decisions of the final model. Of the 2,336 high earners in the test set, it detects 1,493 (recall of 0.639) and misses 843. Among the 7,422 other people, it wrongly predicts >50K for 444. Missed high earners are therefore its main type of error, almost twice as frequent as false alarms. When the model predicts >50K, it is right in 77.1% of cases (1,493 out of 1,937).

10 Fairness Audit

10.1 Setting

We audit the final model by sex (`src/fairness.py`). Sex is a protected attribute, the share of high earners differs strongly between men and women (Section 1), and the model uses it directly (Section 8). The audit uses the final calibrated model and the test set, which contains 3,232 women (10.4% of high earners) and 6,526 men (30.6%). The test set is only used to measure; no choice is made from it. Metrics by group are computed with Fairlearn's `MetricFrame`.

The task does not specify how the model would be used. For the purpose of this exercise, we assume that it identifies people eligible for an offer reserved for incomes above \$50K. The most harmful error is then to miss a person who really earns >50K, since this person is wrongly denied the offer. We therefore choose *equal opportunity*: the true positive rate (TPR) should be the same for men and women.

10.2 Results

	Group	Selection	TPR	FPR	Precision
Before	Women	0.075	0.551	0.020	0.764
	Men	0.260	0.654	0.086	0.772
	Gap	0.185	**0.103**	0.066	0.007
After	Women	0.097	0.631	0.035	0.679
	Men	0.278	0.680	0.100	0.750
	Gap	0.181	**0.049**	0.066	0.071

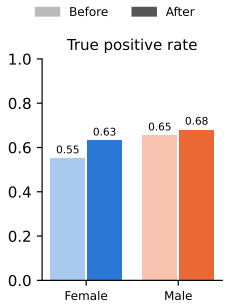


Table 9: Fairness audit by sex on the test set, before mitigation (threshold 0.5) and after mitigation (one threshold per group). Selection is the share of people predicted >50K. The figure shows the TPR of each group.

Before mitigation. With the single threshold of 0.5, the model detects 55.1% of the women who earn >50K, against 65.4% of the men: a gap of 10.3 points in TPR (Table 9). Equal opportunity is therefore not satisfied. In contrast, precision is almost the same for both groups (0.764 and 0.772): when the model predicts >50K, it is equally reliable for women and men.

Mitigation. We use Fairlearn's `ThresholdOptimizer`. It keeps the model unchanged and chooses one decision threshold per group, under the constraint of equal TPR. The thresholds are learned on the training set. Among all the pairs of thresholds that satisfy the constraint, it keeps the one with the best accuracy. Accuracy is not used here to evaluate the model, but only to choose between these pairs. With calibrated probabilities, maximising accuracy corresponds to our reference threshold of 0.5, so the only change comes from the fairness constraint. Indeed, the threshold for men stays close to 0.5 (0.47 to 0.48), while the threshold for women falls to 0.37–0.41.

After mitigation. The TPR gap is halved, from 0.103 to 0.049: the model now detects 63.1% of the female high earners. The gap is not fully closed on the test set, for two reasons: the thresholds were learned on training predictions, which the model saw during training, and the test set contains only about 340 women who earn >50K, so their TPR is estimated with some uncertainty. The cost for the whole test set is small: precision falls from 0.771 to 0.740, recall rises from 0.639 to 0.673, and F1 slightly rises from 0.699 to 0.705. PR-AUC does not change, since the probabilities are the same.

Limits. The mitigation creates another gap: precision becomes lower for women than for men (0.679 against 0.750). When the base rates of two groups differ, a known result states that equal TPR and equal precision cannot both be reached (except by a perfect model). Choosing equal opportunity means accepting this difference in precision. Using a different threshold for each sex is also a policy decision, with legal and ethical implications, which would need to be validated before any real use. The same audit could be carried out for other attributes, such as race.

References

[1] R. Kohavi. Scaling up the accuracy of naive-Bayes classifiers: A decision-tree hybrid. In *Proceedings of KDD*, 1996.

[2] D. B. Rubin. Inference and missing data. *Biometrika*, 63(3):581–592, 1976.

[3] T. Saito and M. Rehmsmeier. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE*, 10(3):e0118432, 2015.

A Additional Figures

Figures 7 to 12 show the results of Sections 4, 6, 9 and 10 as bar charts, one panel per metric.

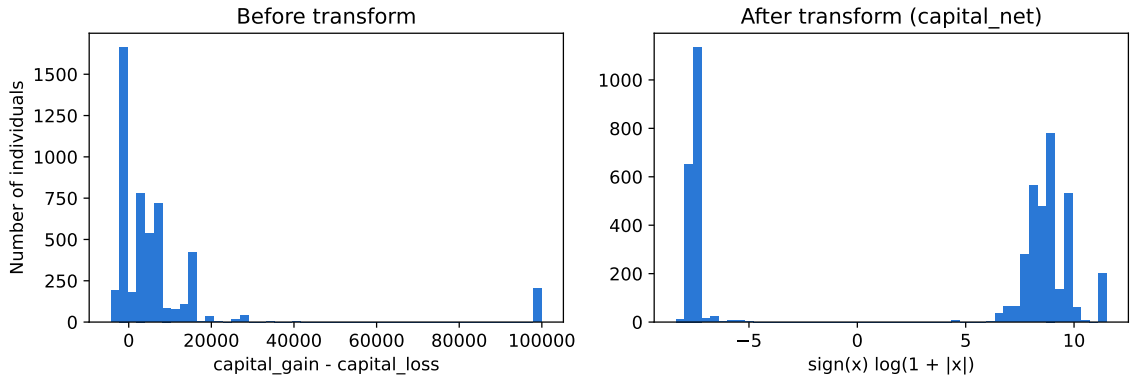


Figure 6: Non-zero net capital income (capital gain minus capital loss, training set) before and after the signed log transform. The transform compresses the long right tail; the two groups after the transform correspond to losses (negative values) and gains.

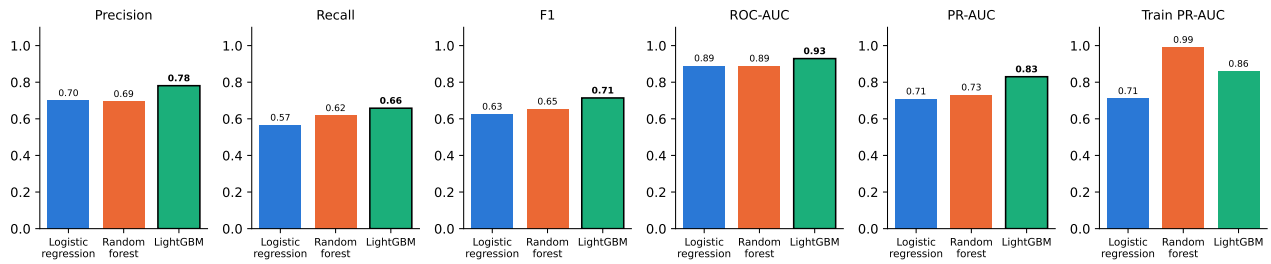


Figure 7: Model comparison with default hyperparameters (Section 4), 5-fold cross-validation. The framed bar is the best model for each metric.

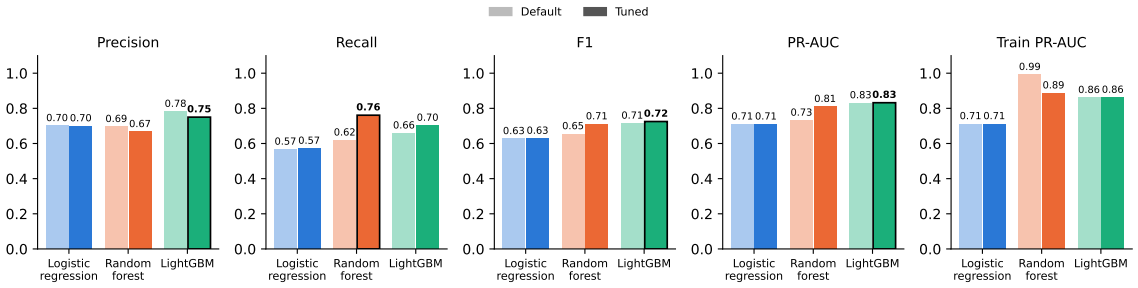


Figure 8: Default (light) and tuned (full colour) models (Section 6), 5-fold cross-validation on the training set, threshold 0.5. The framed bar is the best tuned model for each metric.

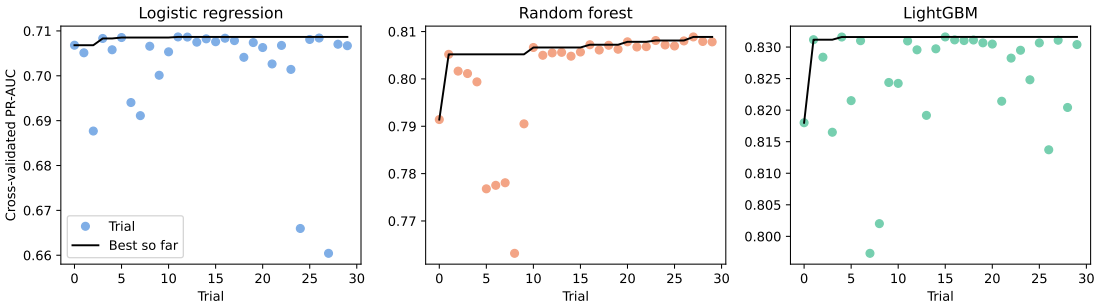


Figure 9: Cross-validated PR-AUC of each Optuna trial (Section 6). The black line is the best value found so far.

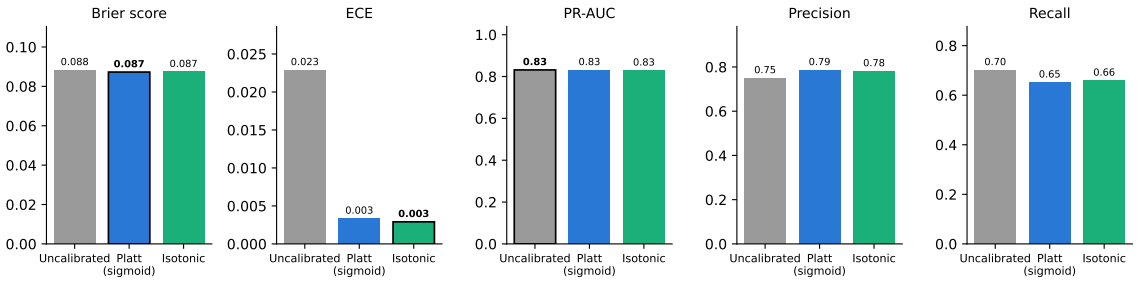


Figure 10: Calibration of the final model (Section 9), 5-fold cross-validation. For the Brier score and the ECE, lower is better.

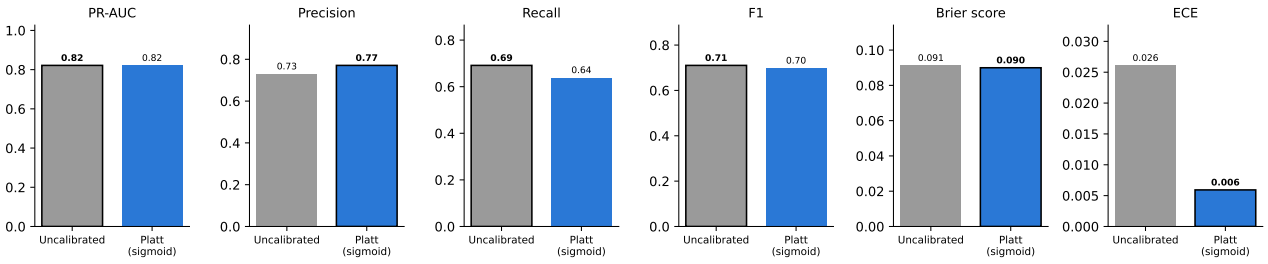


Figure 11: Final evaluation on the test set (Section 9), without and with Platt scaling.

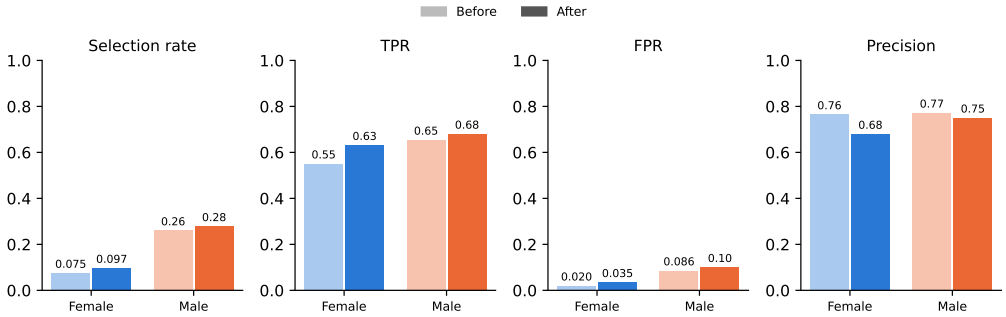


Figure 12: Fairness metrics by sex on the test set, before (light) and after (full colour) mitigation (Section 10).